L-4

CLASS X – SCIENCE

ELECTRICITY

PRASHANT KIRAD

Q. Consider the following combinations of resistors: (1 Mark)

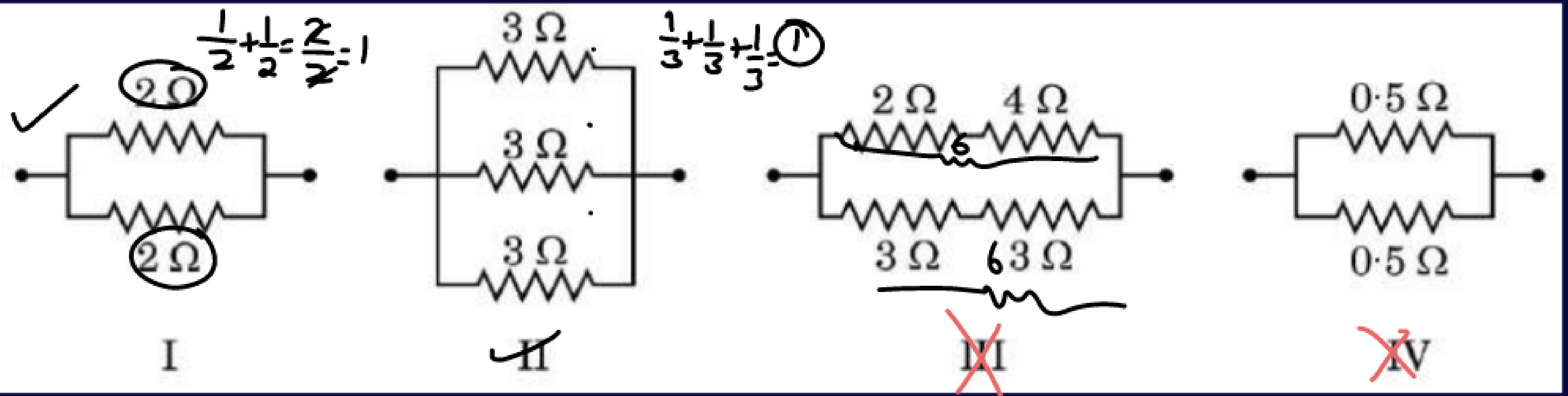
The combinations having equivalent resistance 1 Ω is/are:

(a) I and IV

(b) Only IV

(c) I and II

(d) I, II and III



$\frac{1}{6}+\frac{1}{6}=\frac{2}{6}=\frac{1}{3} \rightarrow 3\Omega$

$\frac{1}{R_{eq}}=\frac{1}{R_1}+\frac{1}{R_2}$

R_{eq}

$\frac{1}{0.5}+\frac{1}{0.5}=\frac{2}{0.5}$

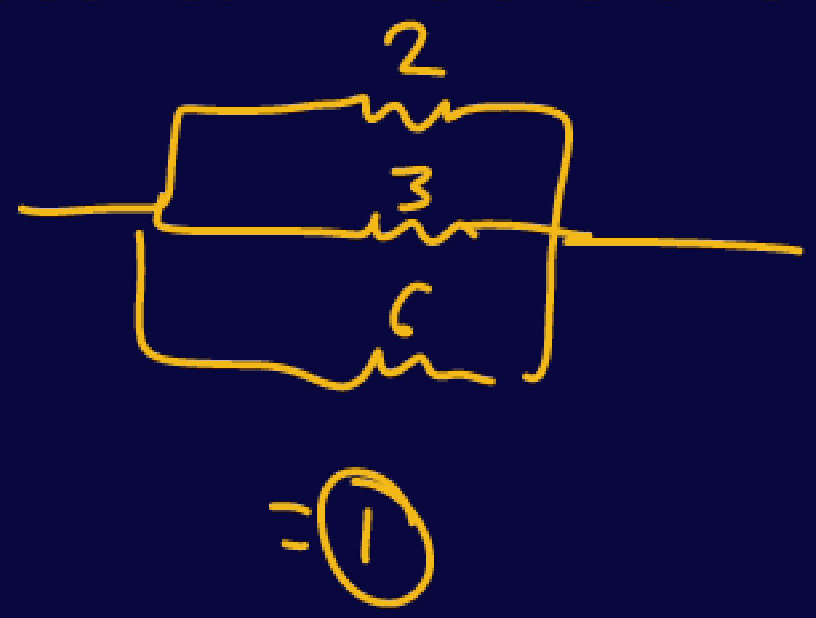
$=\frac{0.5}{2}=0.25\Omega$

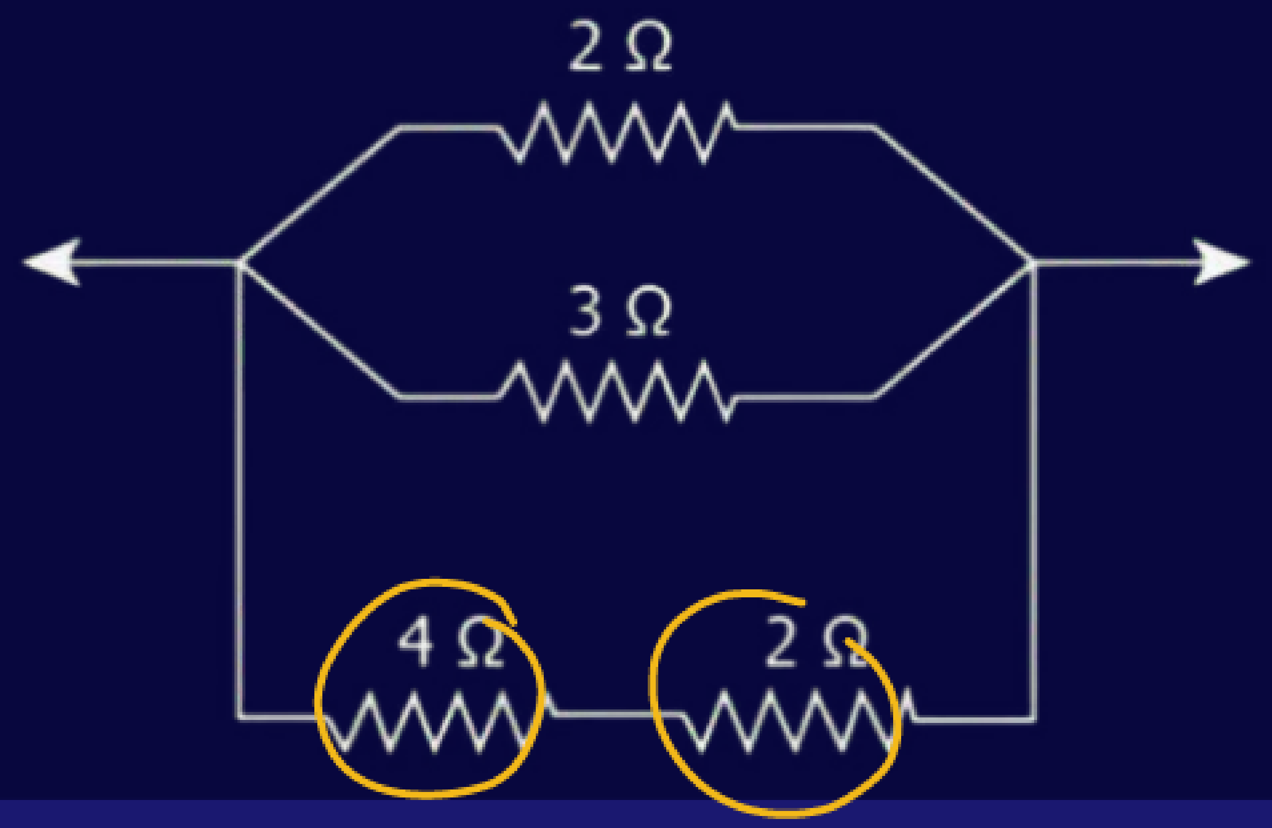
$2\Omega \quad 3\Omega \quad 6\Omega$

$R_{min} \rightarrow \quad \rightarrow 1$

$R_{max} \rightarrow$ Series $\rightarrow 11\Omega$

Q. The image shows a combination of 4 resistors. What is the net resistance between the two points in the circuit?





(a) 0.5 Ω

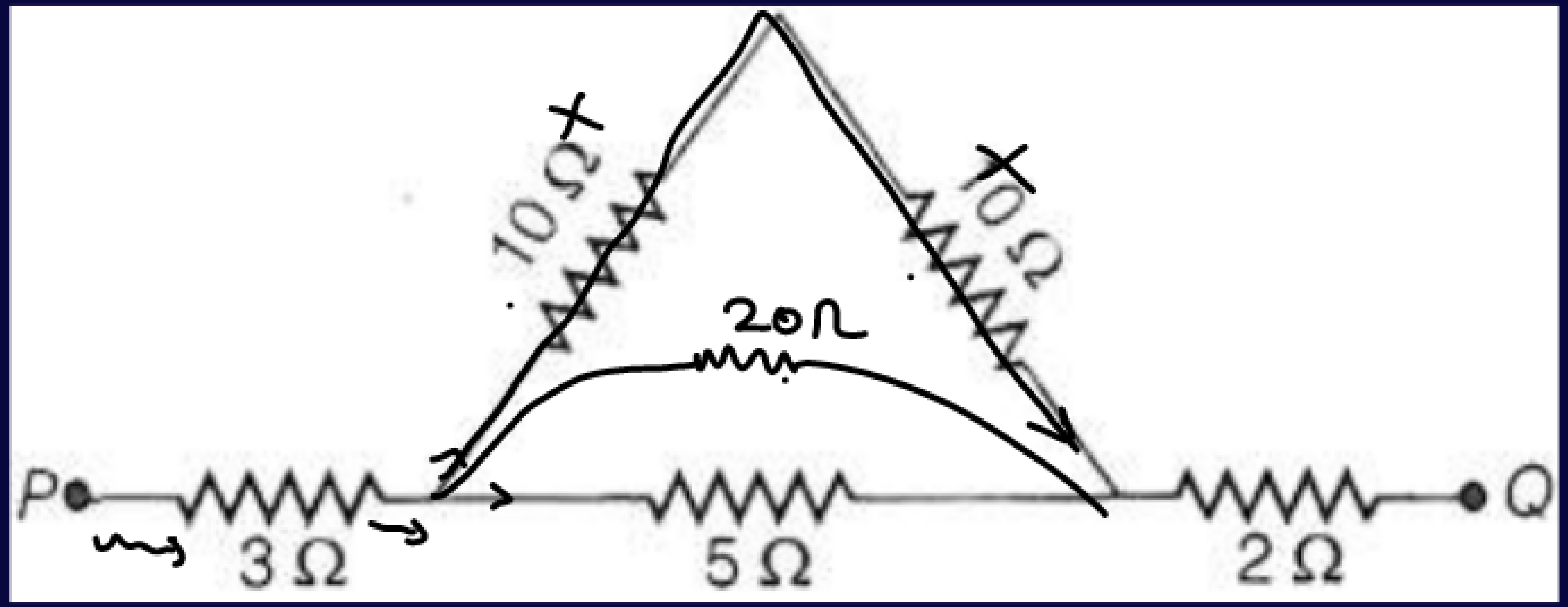
(b) 1.0 Ω

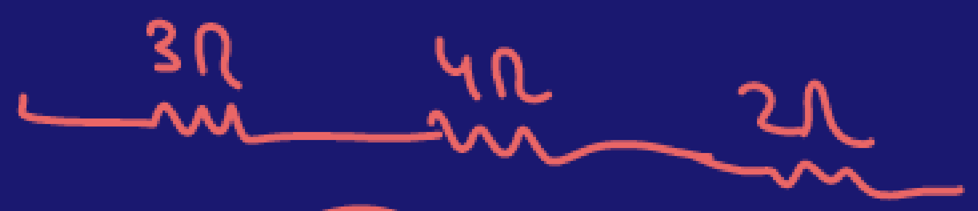
(c) 1.5 Ω

(d) 2.0 Ω

Q. Calculate the equivalent resistance

$\frac{1}{20}+\frac{1}{5}=\frac{4+1}{20}=\frac{5}{20}=\frac{1}{4}\rightarrow 4\Omega$





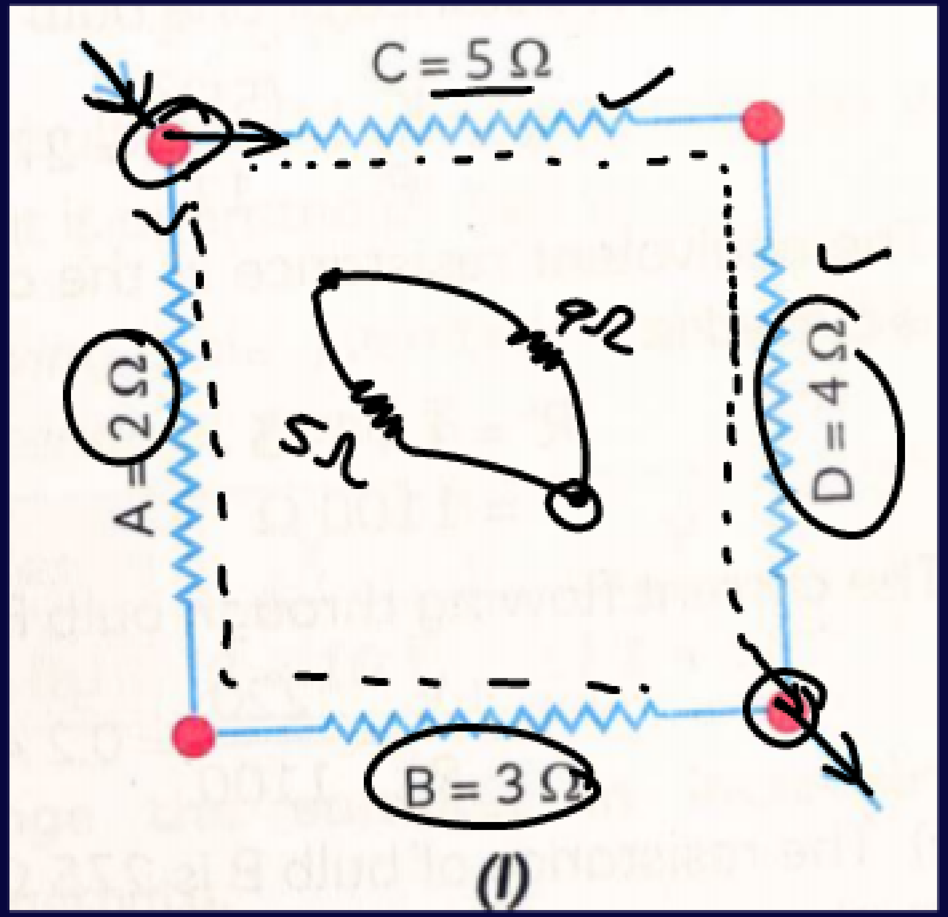
9Ω

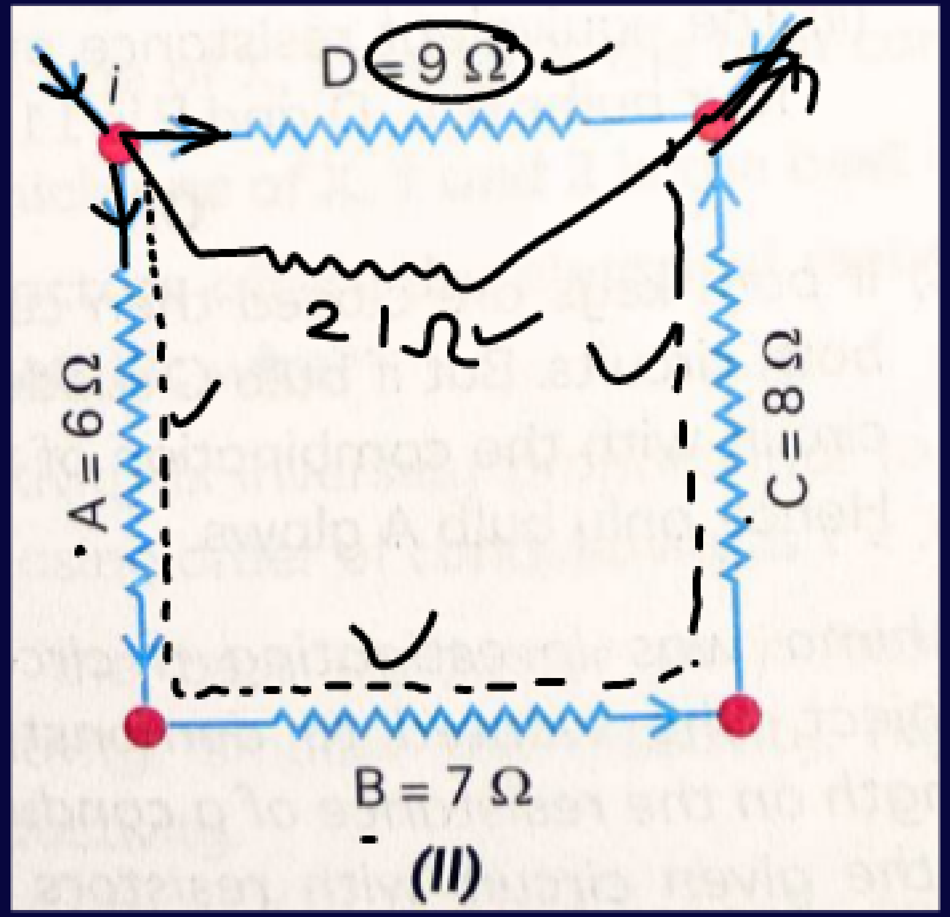
Q. Calculate the total resistance in diagrams (I) and (II).

$$\frac{1}{9}+\frac{1}{21} = \frac{7+3}{63} = \frac{10}{63}$$

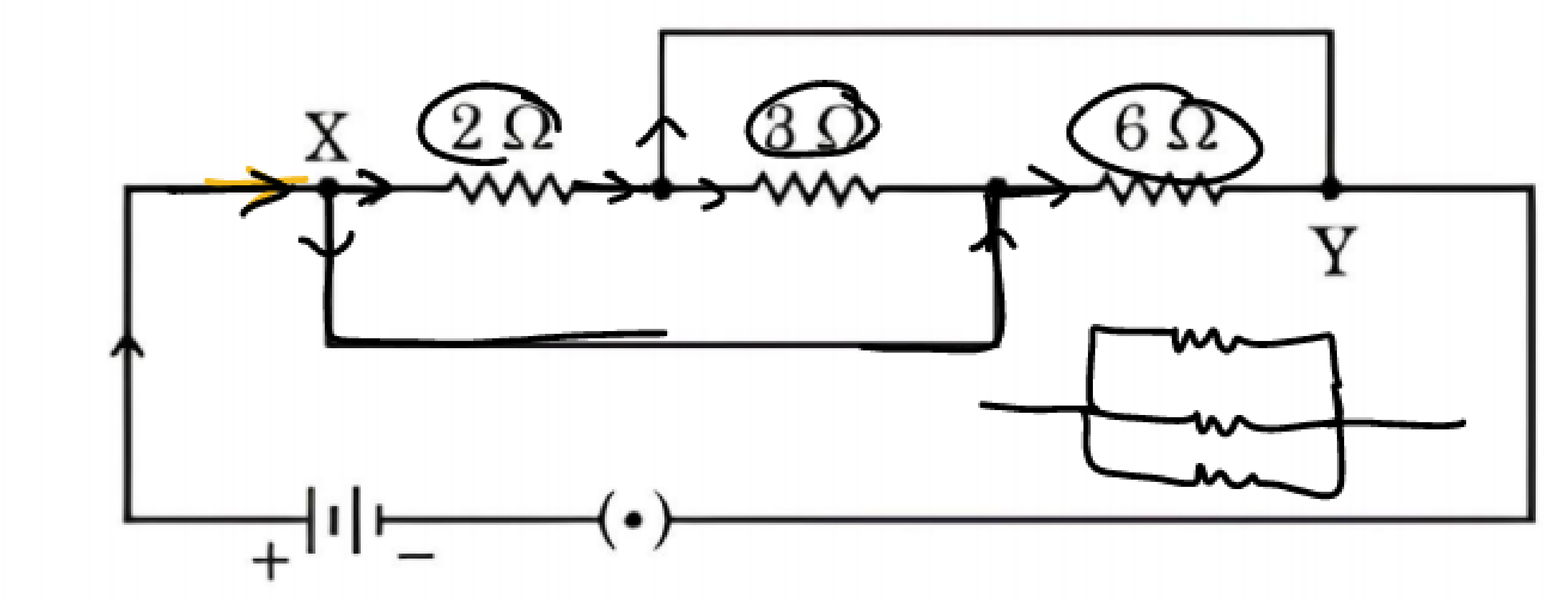
$$\frac{63}{10}\Omega$$

$$\frac{1}{9}+\frac{1}{5} = \frac{14}{45} = \frac{45}{14}\Omega$$





Q. In the given circuit, the total resistance between X and Y is to be calculated. The circuit contains resistors of 2 Ω, 3 Ω and 6 Ω connected as shown in the figure. Find the equivalent resistance between X and Y.



NCERT

Q. A piece of wire of resistance R is cut into five equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R′, then the ratio R/R′ is –

(a) 1/25

(b) 1/5

(c) 5

(d) 25

R/5

$$\frac{1}{R'} = \frac{5}{R} + \frac{5}{R} + \frac{5}{R} + \frac{5}{R} + \frac{5}{R}$$

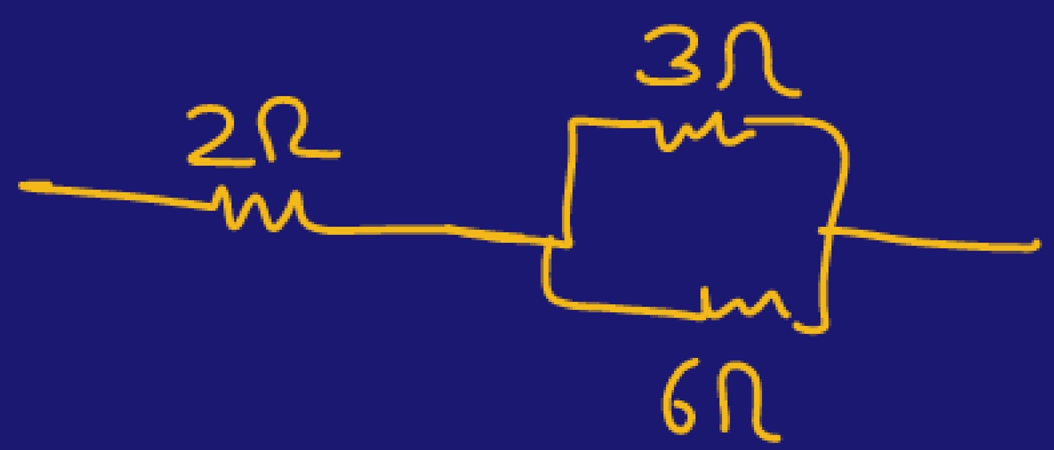
$$= \frac{25}{R} \quad \boxed{\frac{R}{25}}$$

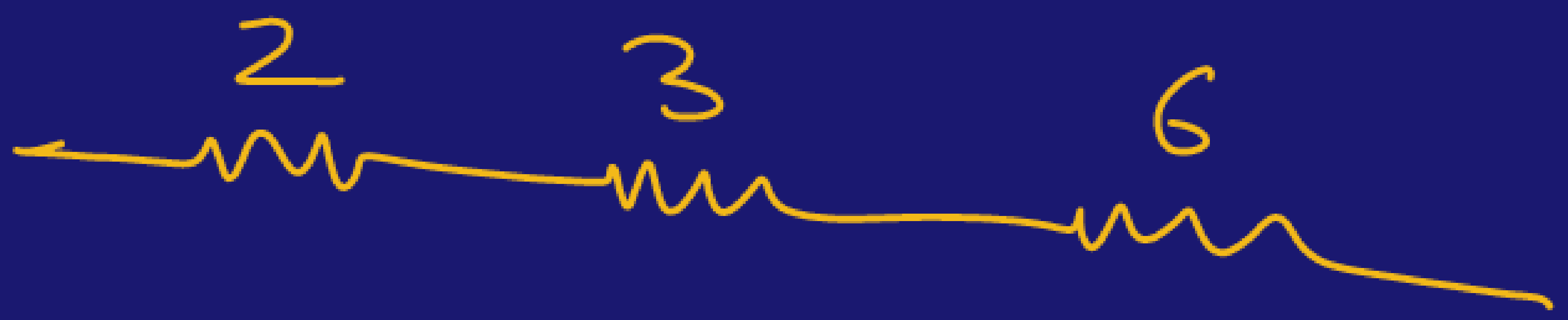
$$R' = \frac{R}{25}$$

$$\frac{R}{\frac{R}{25}} = 25$$

NCERT

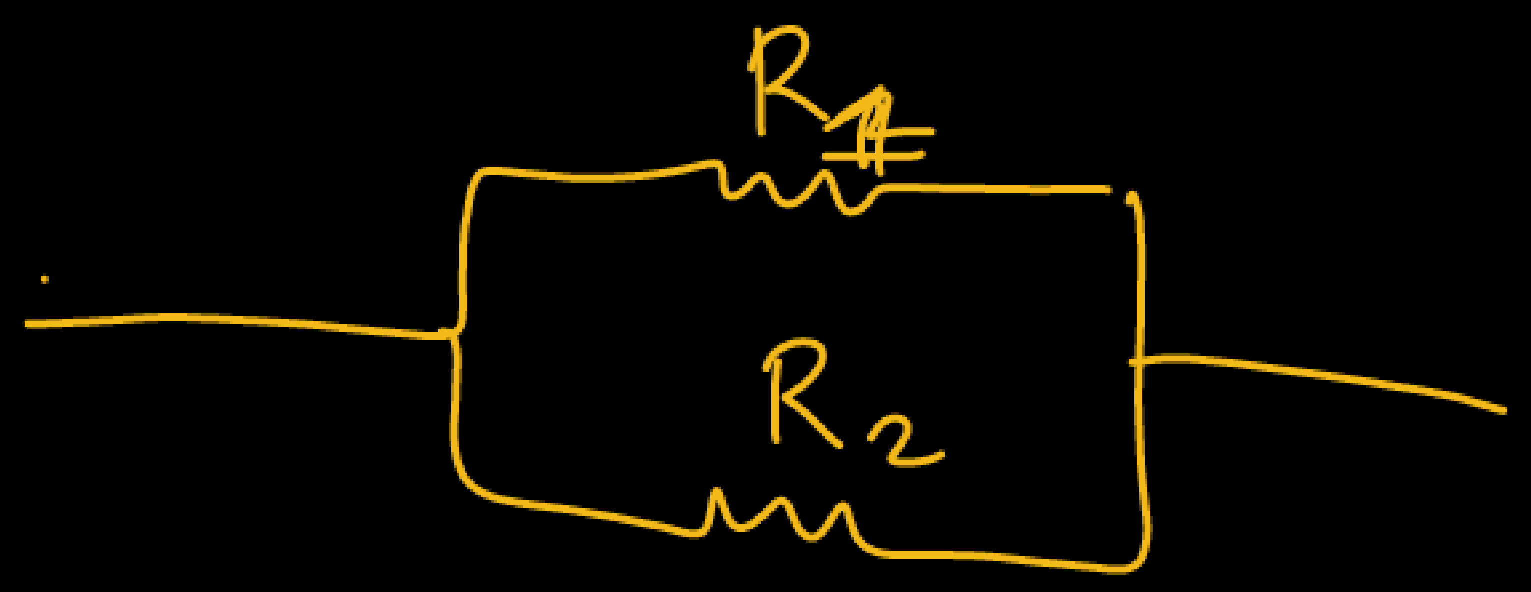
Q. How can three resistors of resistances 2 Ω, 3 Ω, and 6 Ω be connected to give a total resistance of (a) 4 Ω, (b) 1 Ω?





$$R = \frac{3\times 6}{3+6} = \frac{18}{9} = 2\,\Omega$$

PK Trick



$$R_{net} = \frac{R_1 \times R_2}{R_1 + R_2}$$

NCERT

Q. What is (a) the highest, (b) the lowest total resistance that can be secured by combinations of four coils of resistance 4 Ω, 8 Ω, 12 Ω, 24 Ω?

Series

Parallel

4

NCERT

Q. An electric lamp, whose resistance is 20 Ω, and a conductor of 4 Ω resistance are connected to a 6 V battery (Fig. 11.9). Calculate

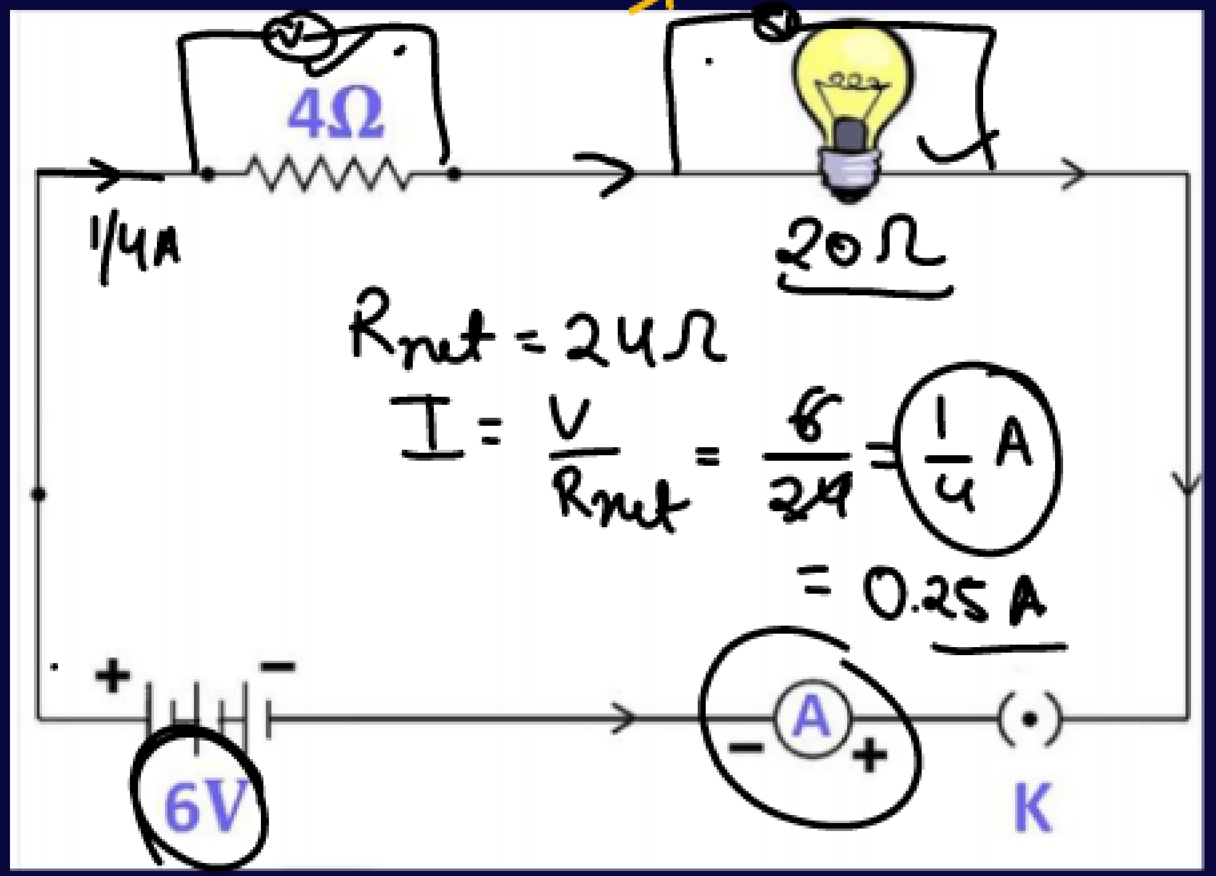
(a) the total resistance of the circuit,

(b) the current through the circuit, and

(c) the potential difference across the electric lamp and conductor.

$V_1 = IR = \frac{1}{4} \times 4 = 1V$

$V = IR = \frac{1}{4} \times 20 = 5V$



1 $\longrightarrow R_{net}$

2 $\longrightarrow I = \dfrac{V}{R_{net}}$

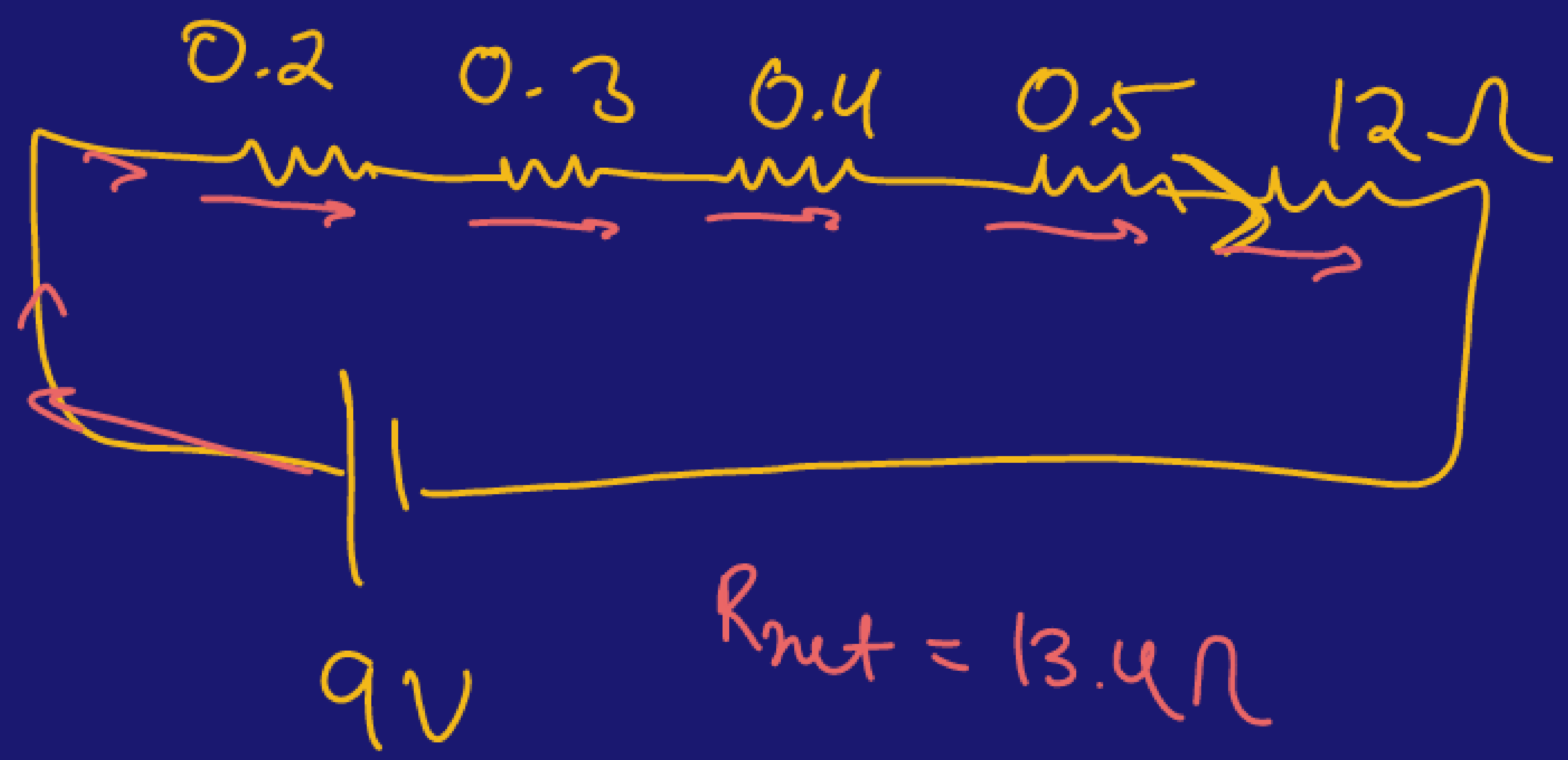
3

NCERT

Q. Draw a schematic circuit diagram consisting of a battery of three cells of 2 V each, a 5 Ω resistor, an 8 Ω resistor, a 12 Ω resistor, a plug key and an ammeter, all connected in series. Also connect a voltmeter to measure the potential difference across the 12 Ω resistor. Calculate the readings of the ammeter and the voltmeter.

NCERT

Q. A battery of 9 V is connected in series with resistors of 0.2 Ω, 0.3 Ω, 0.4 Ω , 0.5 Ω and 12 Ω, respectively. How much current would flow through the 12 Ω resistor?



$R_{net} = 13.4\,\Omega$

$I = \frac{V}{R_{net}} = \frac{9}{13.4}$

NCERT

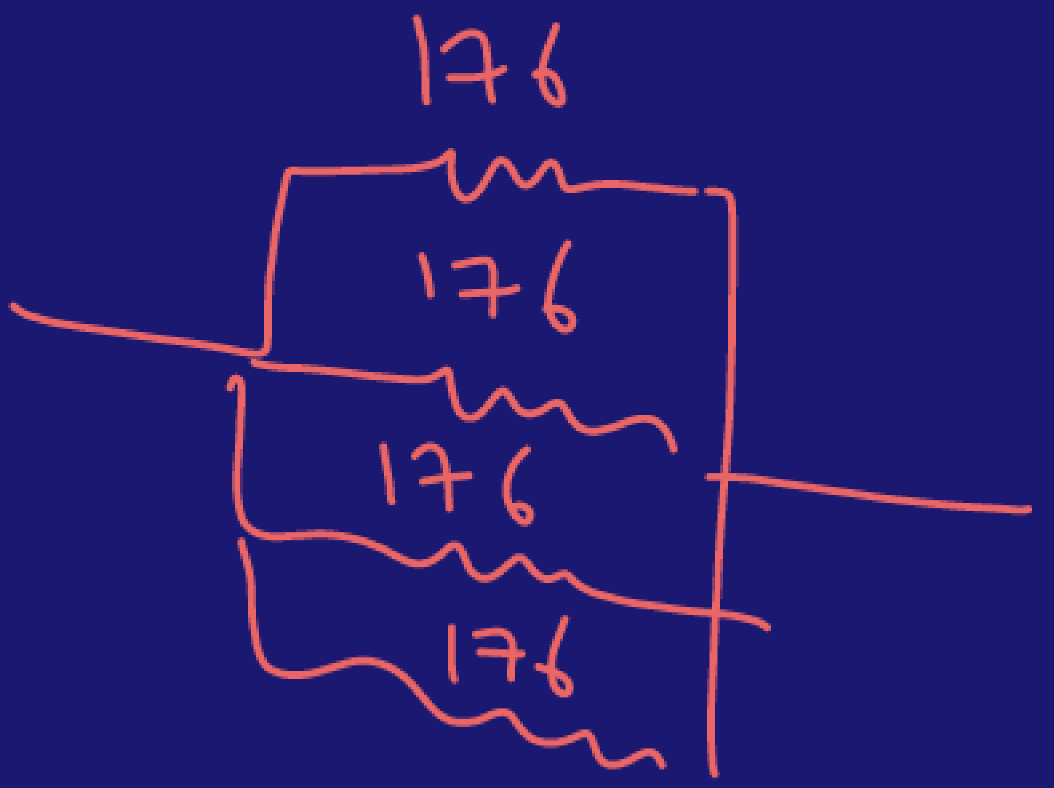
Q. How many 176 Ω resistors (in parallel) are required to carry 5 A on a 220 V line?

$V = 220\,V$

$I = 5\,A$

$V = IR$

$$R = \frac{V}{I} = \frac{\cancel{220}}{\cancel{5}} = 44\,\Omega$$



$$\frac{1}{R_{net}} = \frac{1}{176} + \frac{1}{176} + \frac{1}{176} + \frac{1}{176}$$

$$= \frac{\cancel{4}}{\cancel{176}} = \frac{1}{44} \Rightarrow 44\,\Omega$$

$V = IR = \frac{1}{2} \times 4 = 2V$

Q. In the given circuit, when the key is closed, determine the following:

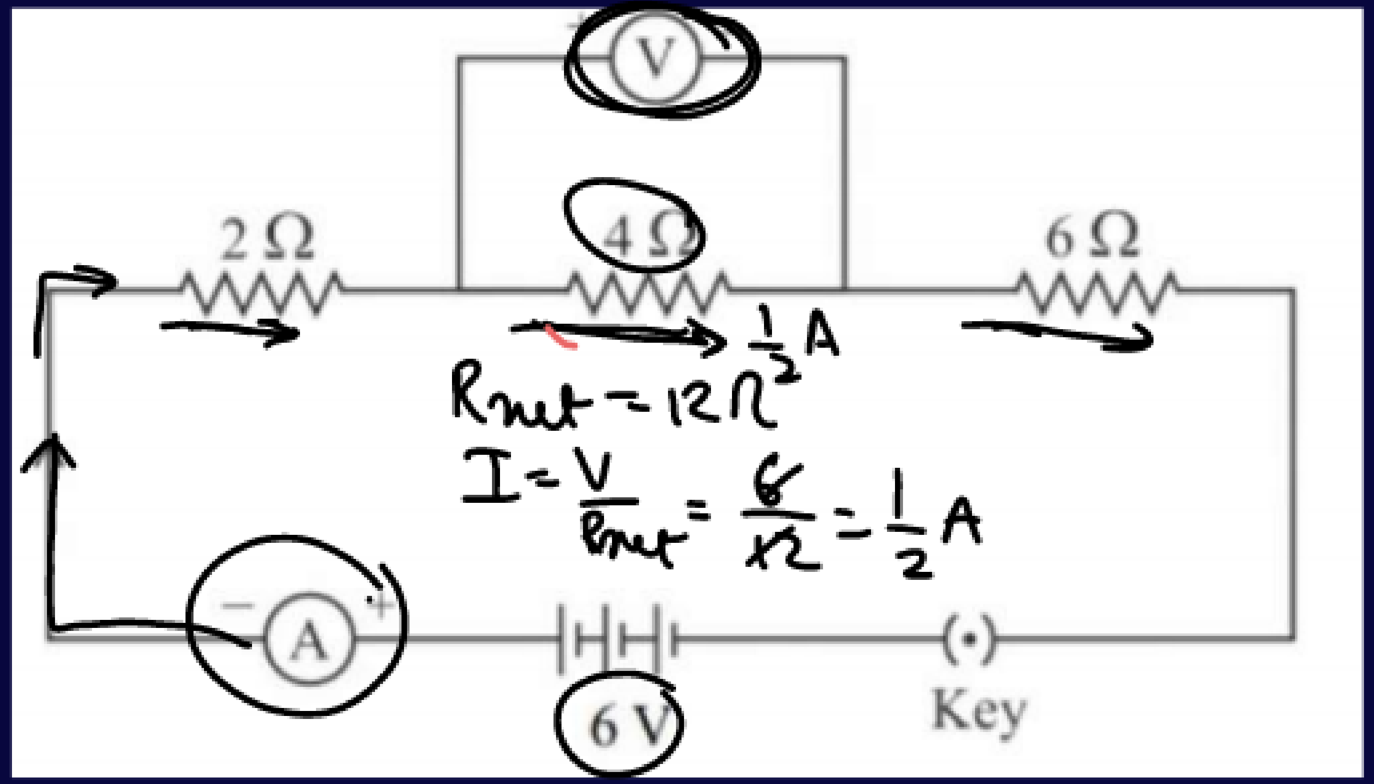
(i) Equivalent resistance of the circuit

(ii) Current flowing in the circuit

(iii) Reading of the ammeter

(iv) Potential difference across the 4 Ω resistor

(v) Reading of the voltmeter



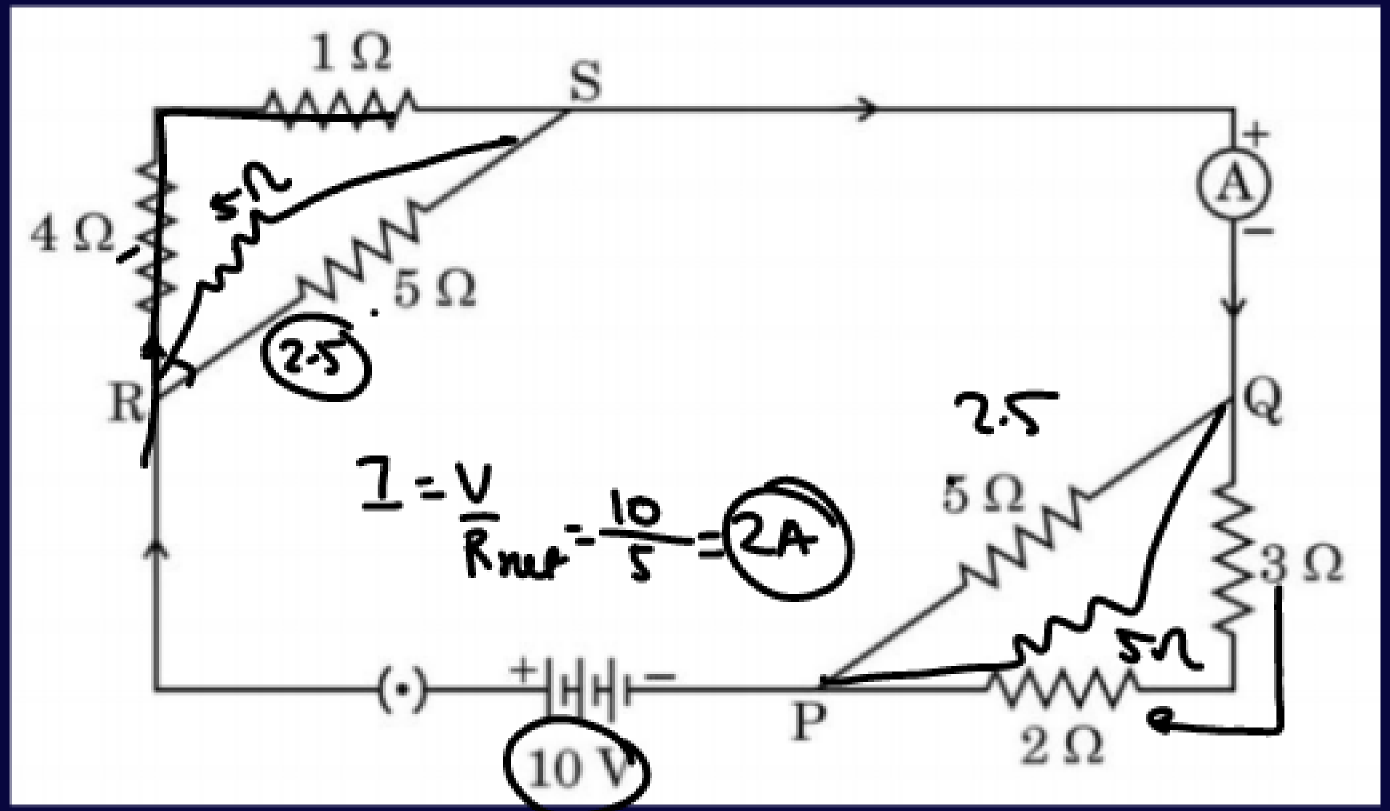
Q. Consider the given electric circuit.

Calculate the following:

(a) Total resistance of the circuit

(b) The electric current drawn from the battery

(c) Potential difference between points P and Q



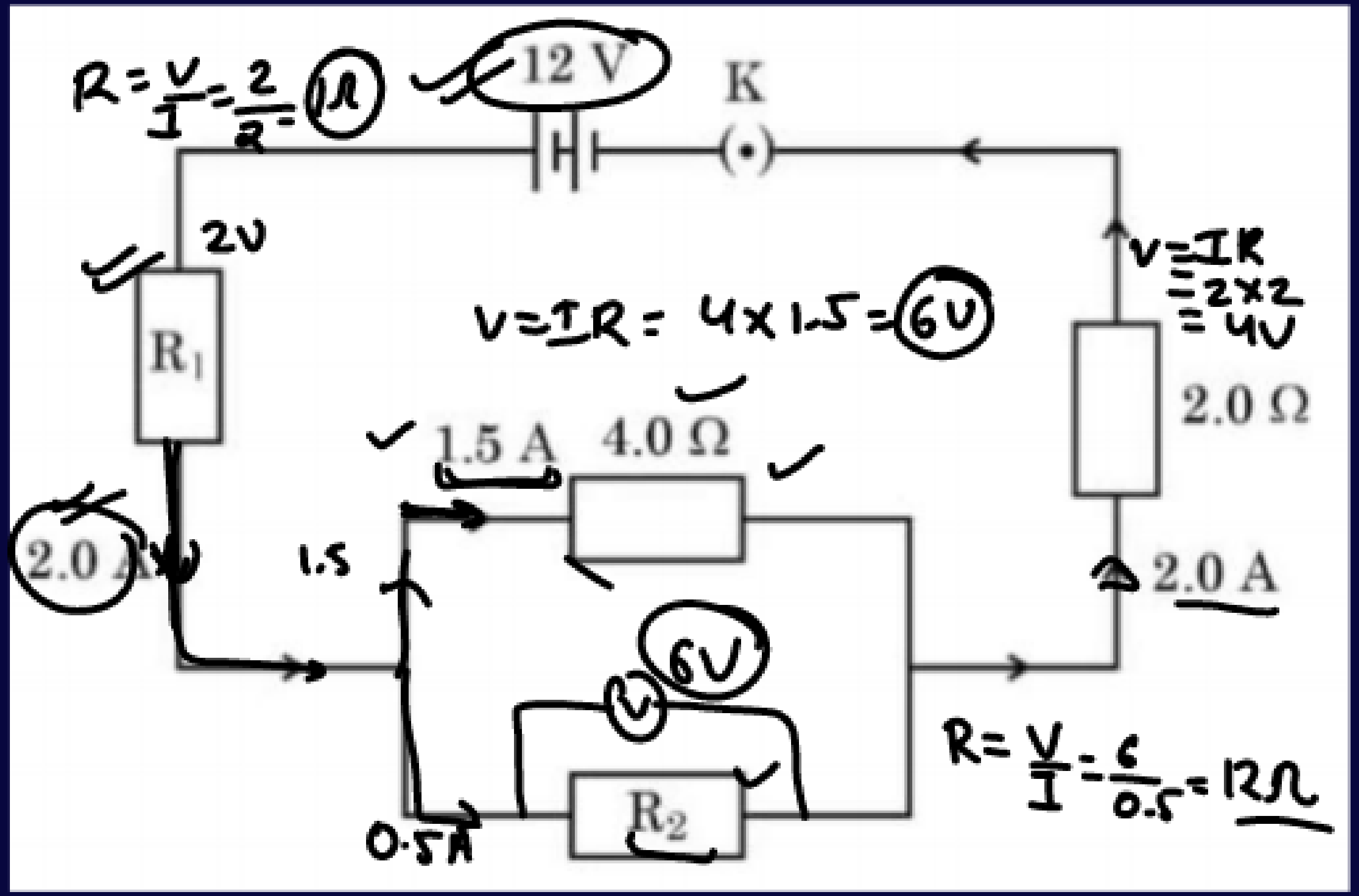
Q. The given electric circuit is a part of an electrical device. Use the information given in the electric circuit diagram to calculate:

(i) Potential difference across the ends of resistor R_2.

(ii) Value of resistor R_2.

(iii) Value of resistor R_1.

Also, write the factors on which resistance of a conductor depends and derive the formula for resistance of a given conductor.



NCERT HW

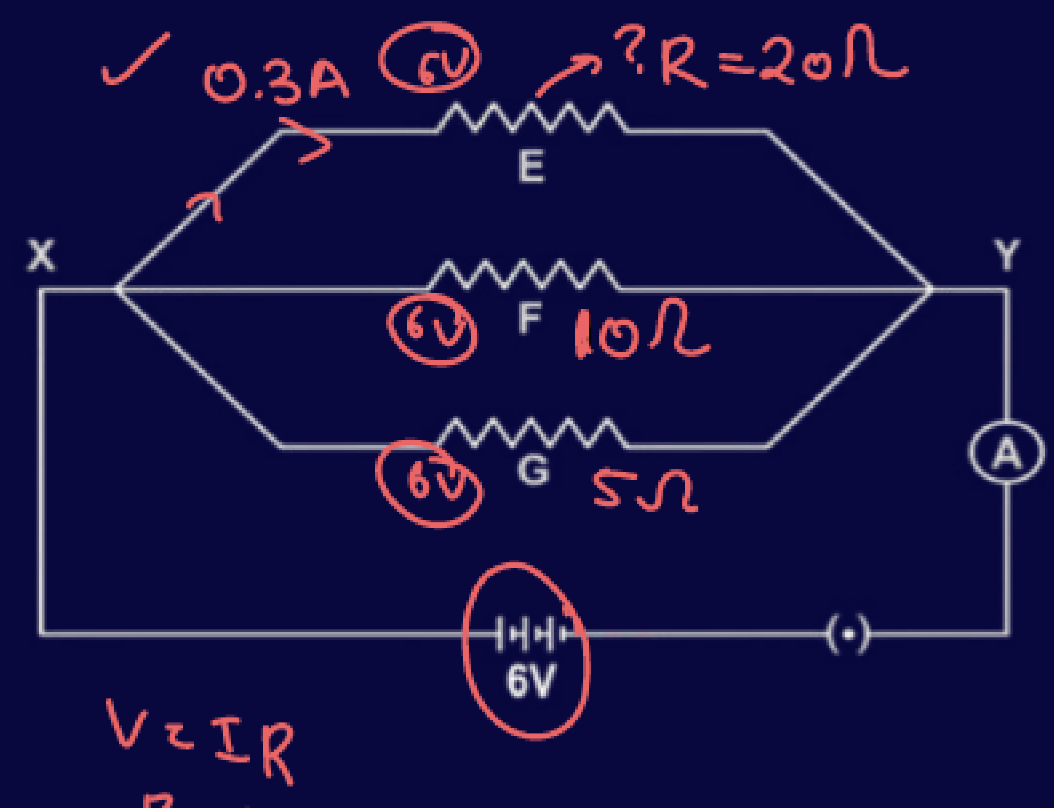
Q. An electric lamp of 100 Ω, a toaster of resistance 50 Ω, and a water filter of resistance 500 Ω are connected in parallel to a 220 V source. What is the resistance of an electric iron connected to the same source that takes as much current as all three appliances, and what is the current through it?

Q. Three resistors in a circuit are attached as shown in the figure. The resistances of F and G are 10 Ω and 5 Ω respectively. The resistance of E is unknown. These resistors are connected to a battery having potential difference 6 V.

(a) What is the term used to describe such an arrangement of resistors?

(b) What is the resistance of E if 0.3 A current flows through it?

(c) What is the total current flowing in the circuit?



$V = IR$

$R = \frac{V}{I} = \frac{6}{0.3} = 20\Omega$

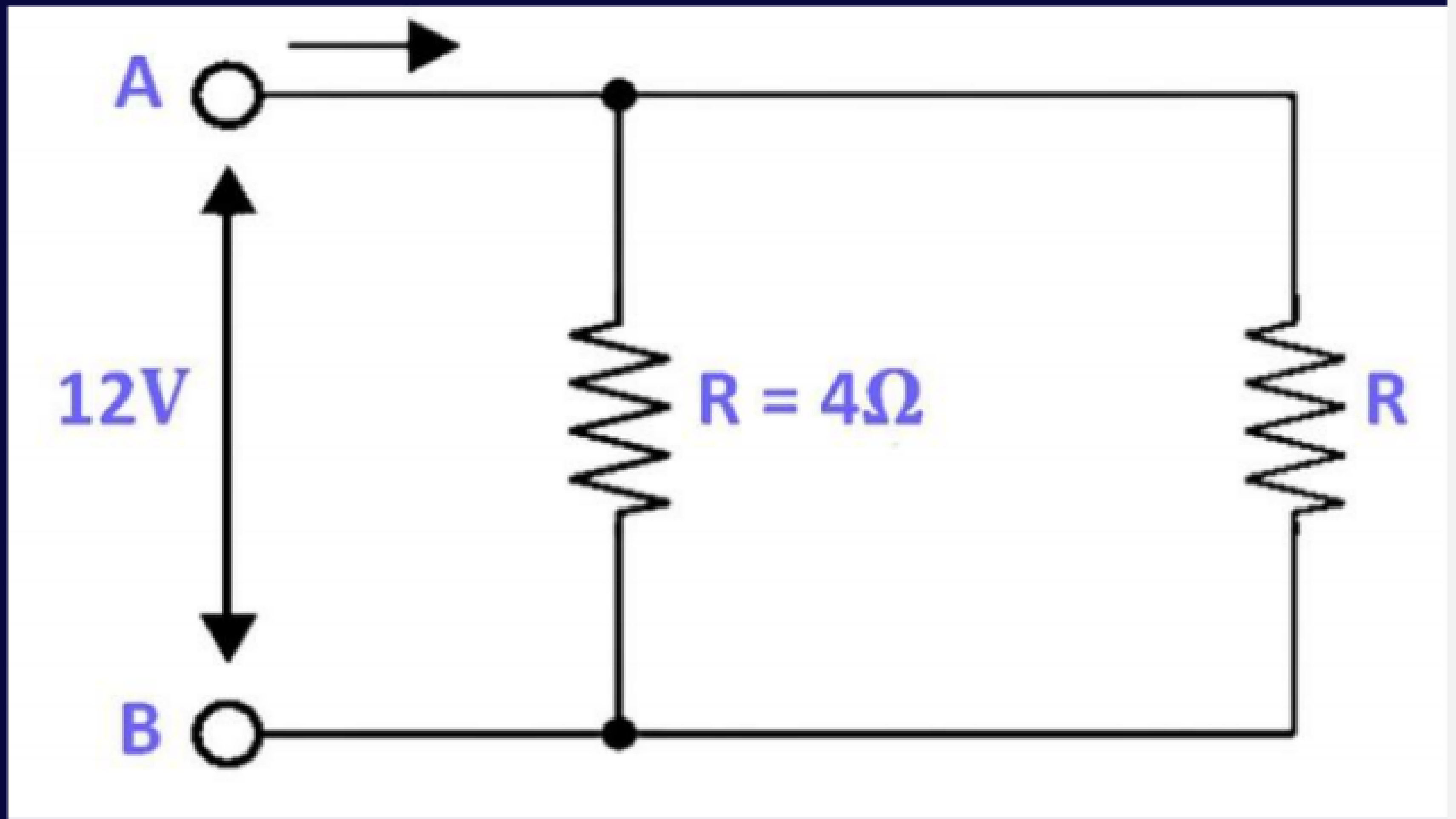
$\frac{1}{R_{net}} = \frac{1}{20} + \frac{1}{10} + \frac{1}{5}$

$= \frac{1+2+4}{20}$

$= \frac{20}{7}\Omega$

$I = \frac{V}{R_{net}} = \frac{6}{20/7} = \frac{21}{10} = 2.1A$

Q. A student has two resistors of 2 Ω and 3 Ω. She has to put one of them in place of R_2, as shown in the circuit. The current needed in the entire circuit is exactly 9 A. Show by calculation which of the two resistors she should choose.



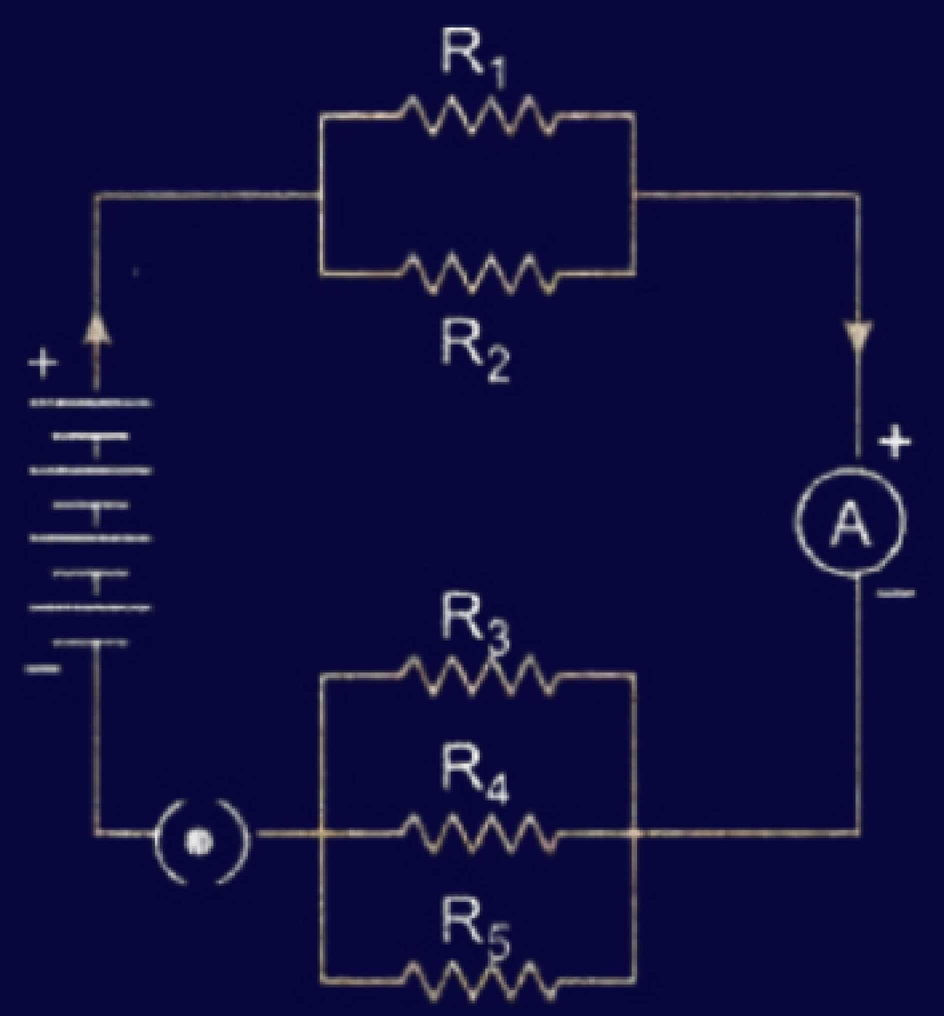
HW

NCERT

Q. If in Fig. 11.12, $R_1 = 10\ \Omega$, $R_2 = 40\ \Omega$, $R_3 = 30\ \Omega$, $R_4 = 20\ \Omega$, $R_5 = 60\ \Omega$, and a 12 V battery is connected to the arrangement. Calculate

(a) the total resistance in the circuit, and

(b) the total current flowing in the circuit.



HW

Q. Find out the following in the electric circuit given in the figure:

(a) Effective resistance of two 8 Ω resistors in the combination

(b) Current flowing through 4 Ω resistor

(c) Potential difference across 4 Ω resistance

(d) Difference in ammeter readings, if any.

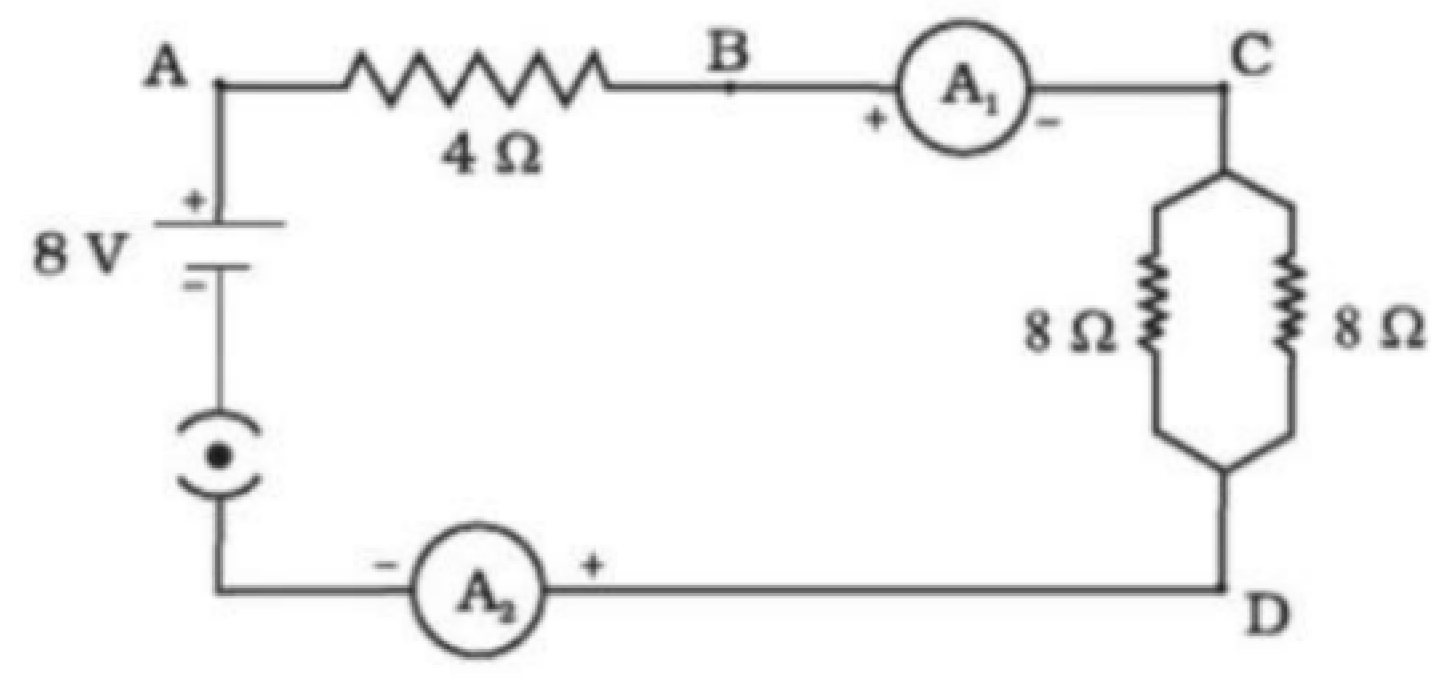


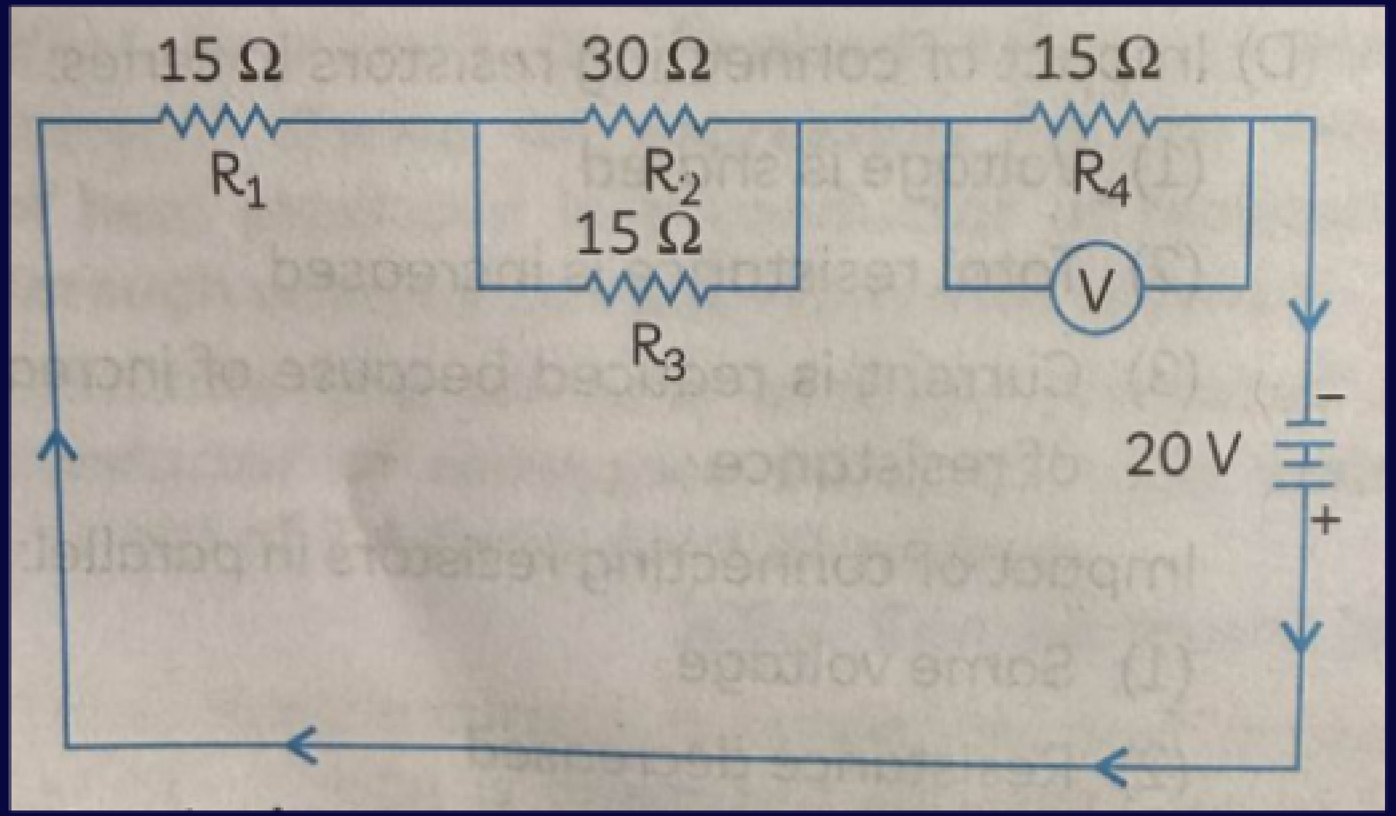
Fig. 12.9

HW

Q. Four resistors, a voltmeter and a battery are connected in a circuit as shown below.

(A) What is the net resistance in the circuit?

(B) How much potential difference will the voltmeter connected across resistor R_4 measure?



HW

Q. The circuit in the figure has resistors in series and parallel.

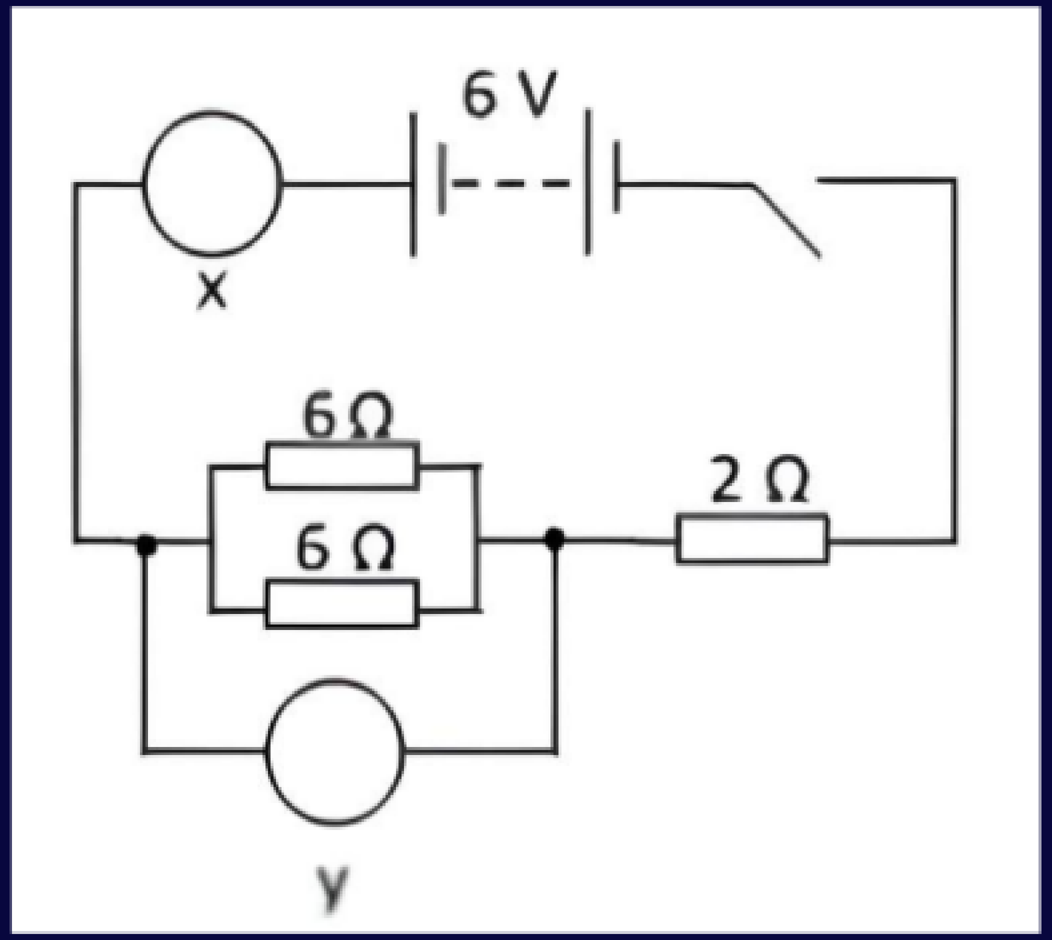
(A) State two factors that affect the resistance of a resistor.

(B) X and Y are meters. Identify X and Y and the electrical property that each meter measures.

(C) (i) Calculate the equivalent resistance in the circuit.

(ii) Using your answer, calculate the current in the circuit. State the unit.

(D) Explain the impact of connecting resistors in series and connecting resistors in parallel on the resistance of a simple circuit.



H.W